

DOT1 analysis continued

Using the original orthofinder file in part 1.

DOT1.1 (not OG0003779) OG0005117 DOT1.2 OG0000629 DOT1.3 OG0000635 DOT1.4 OG0009922 DOT1.5 OG0007957

-> we decided to switch the name of DOT1.1 and DOT1.2, so the final naming is:

DOT1.1 OG0000629 DOT1.2 OG0005117 DOT1.3 OG0000635 DOT1.4 OG0009922 DOT1.5 OG0007957

```
library(tidyverse)
```

Make a list of orthogroup and their corresponding DOT1 subfamilies

```
dot1_og <- tibble(  
  og = c("OG0005117", "OG0000629", "OG0000635", "OG0009922", "OG0007957"),  
  dot1_subfamily = c("DOT1.1", "DOT1.2", "DOT1.3", "DOT1.4", "DOT1.5")  
)
```

Read orthogroup_filt.tsv

```
Orthogroup_filt <-  
  readr::read_tsv("data/Orthogroups_filt.tsv",  
                  col_names = c("Orthogroup", "Ascophyllum_nodosum", "Ath", "X1", "Chordaria_lineage"),  
  dplyr::select(-X1, -X2, -X3) |>  
  dplyr::select(-c(Ascophyllum_nodosum, Ddic, Fser, Saccharina_latissima, Saccorhiza_polyschides,  
Orthogroup_filt.long <- Orthogroup_filt |>  
  tidyr::pivot_longer(-Orthogroup, names_to = "species", values_to = "gene") |>  
  # make new row for each gene in the ", " separated list  
  tidyr::separate_rows(gene, sep = ", ") |>  
  tidyr::drop_na(gene) |>  
  dplyr::mutate(gene = stringr::str_replace(gene, "\\+", "_")) |>  
  dplyr::mutate(species_short = stringr::str_remove(species, ".*$")) |>  
  dplyr::mutate(species_short = dplyr::case_when(  
    species_short == "Cyanidioschyzon" ~ "Cyani",  
    species_short == "Chlamydomonas" ~ "Chlamy",  
    TRUE ~ species_short  
  )) |>  
  dplyr::mutate(clade = dplyr::case_when(  
    species_short %in% c("Esp7", "Ddud", "Dher", "Upin", "Spro") ~ "phaeophyceae",  
    species_short %in% c("Sisc", "Tribonema", "Chrysoparadoxa", "Heterosigma") ~ "early ochrophyt",  
    species_short %in% c("Hsap", "Mmus", "Drme") ~ "animalia",  
    species_short %in% c("Ath", "Oryza", "Marchantia", "Chlamy") ~ "green lineage",  
    species_short %in% c("Cyani") ~ "red algae",  
    species_short %in% c("Yeast") ~ "fungi",  
    species_short %in% c("Fistulifera", "Phaeodactylum") ~ "diatoms",  
    species_short %in% "Cyanophora" ~ "glaucomphyta",  
    TRUE ~ "others"  
  )) |>  
  # dplyr::mutate(  
  )
```

```

#           gene_short = stringr::str_remove(gene, "^.*\\|/")
# ) |>
# remove "contig" (case insensitive), "prot_", any capital letters after the last "_", "ISG/",
# remove "Ddudresnayi_", "C-australica_", "D-herbacea_M_", "H-akashiwo_", "S-ischiensis_", "S-
dplyr::mutate(
  gene_short = gene |>
    stringr::str_replace_all("(?i)contig", "") |>
    stringr::str_replace_all("prot_", "") |>
    stringr::str_replace_all("ISG\\|", "") |>
    stringr::str_replace_all("sp\\|", "") |>
    stringr::str_replace_all("tr\\|", "") |>
    stringr::str_replace_all("Trimin1\\|", "") |>
    stringr::str_replace_all("\\|len_.*$", "") |>
    stringr::str_replace_all("fgenes1_p", "") |>
    stringr::str_replace_all("_([A-Z]+)$", "") |>
    stringr::str_replace_all("_CYAM1$", "")
) |>
dplyr::mutate(
  gene_short = gene_short |>
    stringr::str_replace_all("Ddudresnayi_", "") |>
    stringr::str_replace_all("C-australica_", "") |>
    stringr::str_replace_all("D-herbacea_M_", "") |>
    stringr::str_replace_all("H-akashiwo_", "") |>
    stringr::str_replace_all("S-ischiensis_", "") |>
    stringr::str_replace_all("S-promiscuus_M_", "")
) |>
# remove any repeated sequence after "|", i.e. "A0A1Z5K627|A0A1Z5K627"
dplyr::mutate(
  gene_short = stringr::str_replace(gene_short, "^(.*)\\|\\|1$", "\\|1")
) |>
# remove everything before "Ctg"
dplyr::mutate(
  gene_short = stringr::str_replace(gene_short, "^.*(Ctg.*)$", "\\|1")
)
# Orthogroup_filt.long |>
#   dplyr::pull(gene) |>
#   readr::write_lines("data/dot1_genes_v2.txt")

```

Read in newick file and use ggtree to visualize

```

library(ggtree)
file_OG0005117 <- "data/genetree/OG0005117.filt4.aln.clipkit.treefile"
file_OG0000629 <- "data/genetree/OG0000629.filt4.aln.clipkit.treefile"
file_OG0000635 <- "data/genetree/OG0000635.filt4.aln.clipkit.treefile"
file_OG0009922 <- "data/genetree/OG0009922.filt4.aln.clipkit.treefile"
file_OG0007957 <- "data/genetree/OG0007957.filt4.aln.clipkit.treefile"

tree_OG0005117 <- ggtree::read.tree(file_OG0005117)
tree_OG0000629 <- ggtree::read.tree(file_OG0000629)
tree_OG0000635 <- ggtree::read.tree(file_OG0000635)
tree_OG0009922 <- ggtree::read.tree(file_OG0009922)
tree_OG0007957 <- ggtree::read.tree(file_OG0007957)

# # filter tip labels to only those in Orthogroup_filt.long

```

```

# tree_OG0005117 <- treeio::drop.tip(tree_OG0005117,
#                                   setdiff(tree_OG0005117$tip.label,
#                                           Orthogroup_filt.long$gene))
# tree_OG0000629 <- treeio::drop.tip(tree_OG0000629,
#                                   setdiff(tree_OG0000629$tip.label,
#                                           Orthogroup_filt.long$gene))
# tree_OG0000635 <- treeio::drop.tip(tree_OG0000635,
#                                   setdiff(tree_OG0000635$tip.label,
#                                           Orthogroup_filt.long$gene))
# tree_OG0009922 <- treeio::drop.tip(tree_OG0009922,
#                                   setdiff(tree_OG0009922$tip.label,
#                                           Orthogroup_filt.long$gene))
# tree_OG0007957 <- treeio::drop.tip(tree_OG0007957,
#                                   setdiff(tree_OG0007957$tip.label,
#                                           Orthogroup_filt.long$gene))

# # rename tip label to species via Orthogroup_filt.long
# tree_OG0005117$tip.label <-
#   Orthogroup_filt.long$species_short[match(tree_OG0005117$tip.label,
#                                             Orthogroup_filt.long$gene)]
# tree_OG0000629$tip.label <-
#   Orthogroup_filt.long$species_short[match(tree_OG0000629$tip.label,
#                                             Orthogroup_filt.long$gene)]
# tree_OG0000635$tip.label <-
#   Orthogroup_filt.long$species_short[match(tree_OG0000635$tip.label,
#                                             Orthogroup_filt.long$gene)]
# tree_OG0009922$tip.label <-
#   Orthogroup_filt.long$species_short[match(tree_OG0009922$tip.label,
#                                             Orthogroup_filt.long$gene)]
# tree_OG0007957$tip.label <-
#   Orthogroup_filt.long$species_short[match(tree_OG0007957$tip.label,
#                                             Orthogroup_filt.long$gene)]

```

Plot trees via daylight method

```

# "#1B9E77" "#D95F02" "#7570B3" "#E7298A" "#66A61E" "#E6AB02" "#A6761D" "gray40"

```

```

dd = as.data.frame(Orthogroup_filt.long |> dplyr::distinct(gene, gene_short, species, species_short, clade))
# set consistent colour for clade points and labels

```

```

colour_code <- c(
  "early ochrophytina" = "#D95F02",
  "phaeophyceae" = "#A6761D",
  "animalia" = "#7570b3",
  "red algae" = "darkred",
  "green lineage" = "#66a61e",
  "fungi" = "#e6ab02",
  "diatoms" = "gray40",
  "glaucophyta" = "#1b9e77"
)

```

```

# colour by species_short
p1 <- ggtree::ggtree(tree_OG0005117, layout="daylight") %<+% dd +
  geom_tippoint(aes(colour = clade, label = clade),

```

```

        size=2) +
    geom_tiplab(aes(fill = clade,
                    label = paste0(species_short)),
               # label = paste0(species_short, "\n", gene_short)),
               size=1.5,
               color = "black", # color for label font
               geom = "text",
               show.legend = FALSE) +
    ggplot2::scale_colour_manual(values = colour_code) +
    geom_treescale(width = 0.5)

p2 <- ggtree::ggtree(tree_OG0000629, layout="daylight") %<+% dd +
    geom_tippoint(aes(colour = clade, label = clade),
                 size=2) +
    geom_tiplab(aes(fill = clade,
                    label = paste0(species_short)),
               # label = paste0(species_short, gene_short)),
               size=1.5,
               color = "black", # color for label font
               geom = "text",
               show.legend = FALSE) +
    ggplot2::scale_colour_manual(values = colour_code) +
    geom_treescale(width = 0.5)

p3 <- ggtree::ggtree(tree_OG0000635, layout="daylight") %<+% dd +
    geom_tippoint(aes(colour = clade, label = clade),
                 size=2) +
    geom_tiplab(aes(fill = clade,
                    label = paste0(species_short)),
               # label = paste0(species_short, "\n", gene_short)),
               size=1.5,
               color = "black", # color for label font
               geom = "text",
               show.legend = FALSE) +
    ggplot2::scale_colour_manual(values = colour_code) +
    geom_treescale(width = 0.5)

p4 <- ggtree::ggtree(tree_OG0009922, layout="daylight") %<+% dd +
    geom_tippoint(aes(colour = clade, label = clade),
                 size=2) +
    geom_tiplab(aes(fill = clade,
                    label = paste0(species_short)),
               # label = paste0(species_short, "\n", gene_short)),
               size=1.5,
               color = "black", # color for label font
               geom = "text",
               show.legend = FALSE) +
    ggplot2::scale_colour_manual(values = colour_code) +
    geom_treescale(width = 0.5)
    # geom_tiplab(aes(fill = clade,
p5 <- ggtree::ggtree(tree_OG0007957, layout="daylight") %<+% dd +
    geom_tippoint(aes(colour = clade, label = clade),
                 size=2) +
    geom_tiplab(aes(fill = clade,
                    label = paste0(species_short)),

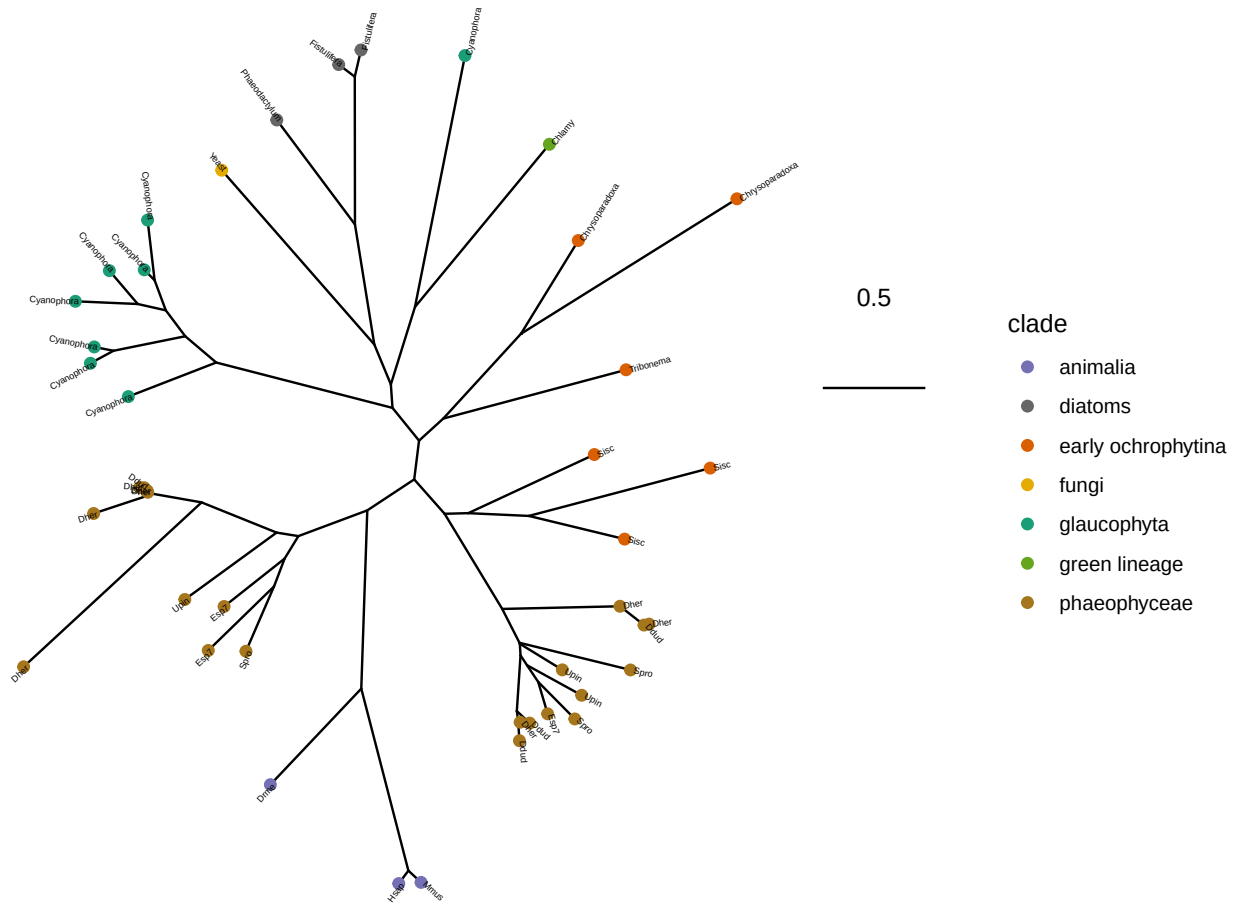
```

DOT1.1 (not OG0003779) OG0005117 DOT1.2 OG0000629 DOT1.3 OG0000635 DOT1.4 OG0009922 DOT1.5
OG0007957

DOT1.1 (OG0005117)

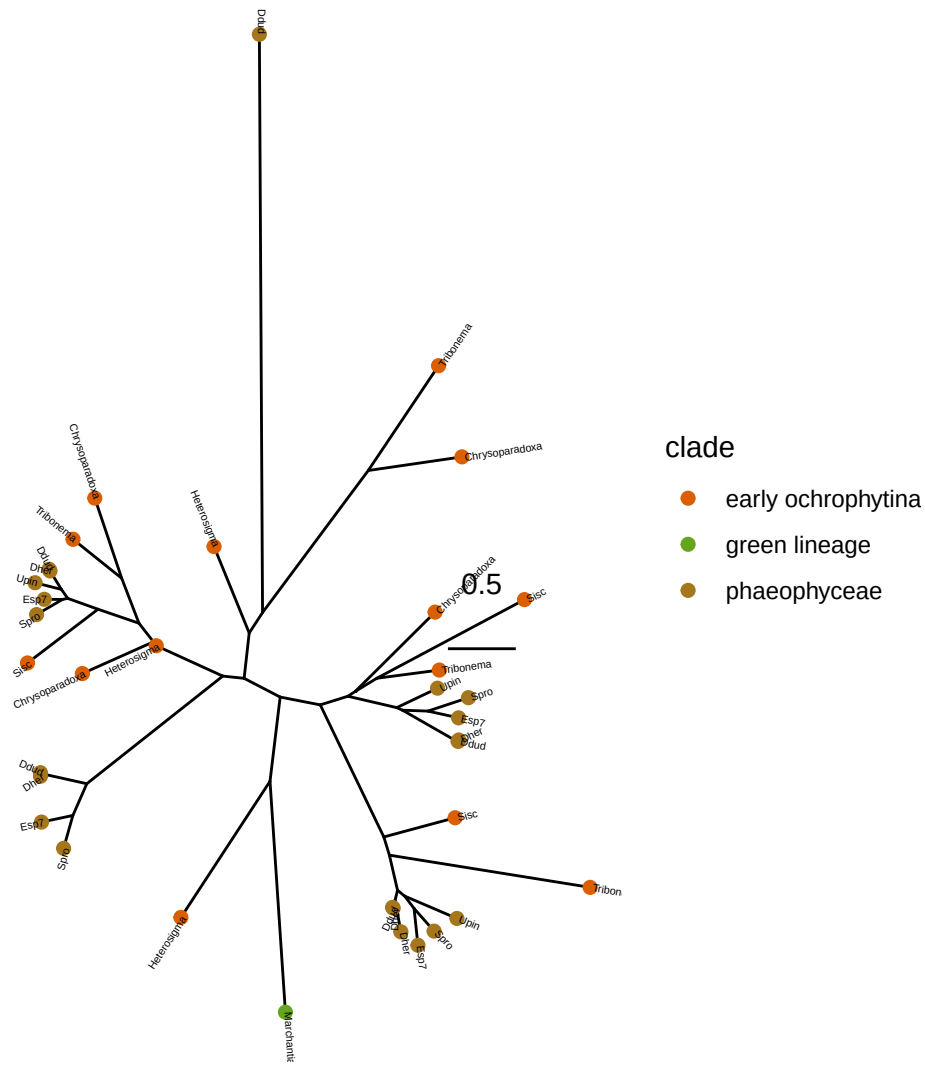


DOT1.2 (OG0000629)



```
# ggplot2::ggsave("figure/DOT1.2_tree_filt4.pdf", width = 8, height = 4)
p3 + ggplot2::labs(title = "DOT1.3 (OG0000635)")
```

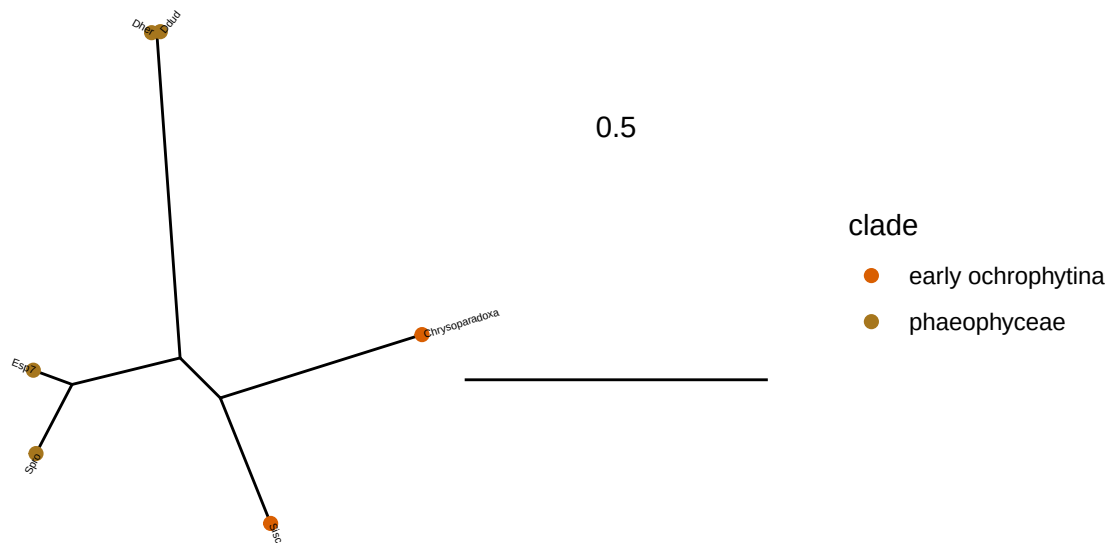
DOT1.3 (OG0000635)



```
# ggplot2::ggsave("figure/DOT1.3_tree_fil3.pdf", width = 8, height = 6)
```

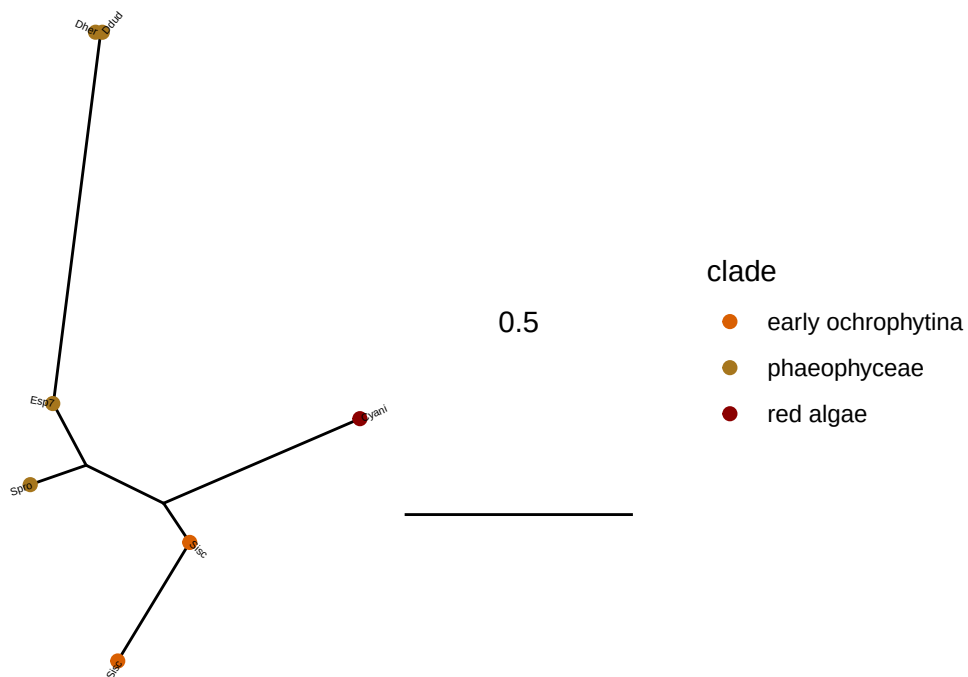
```
p4 + ggplot2::labs(title = "DOT1.4 (OG0009922)")
```

DOT1.4 (OG0009922)



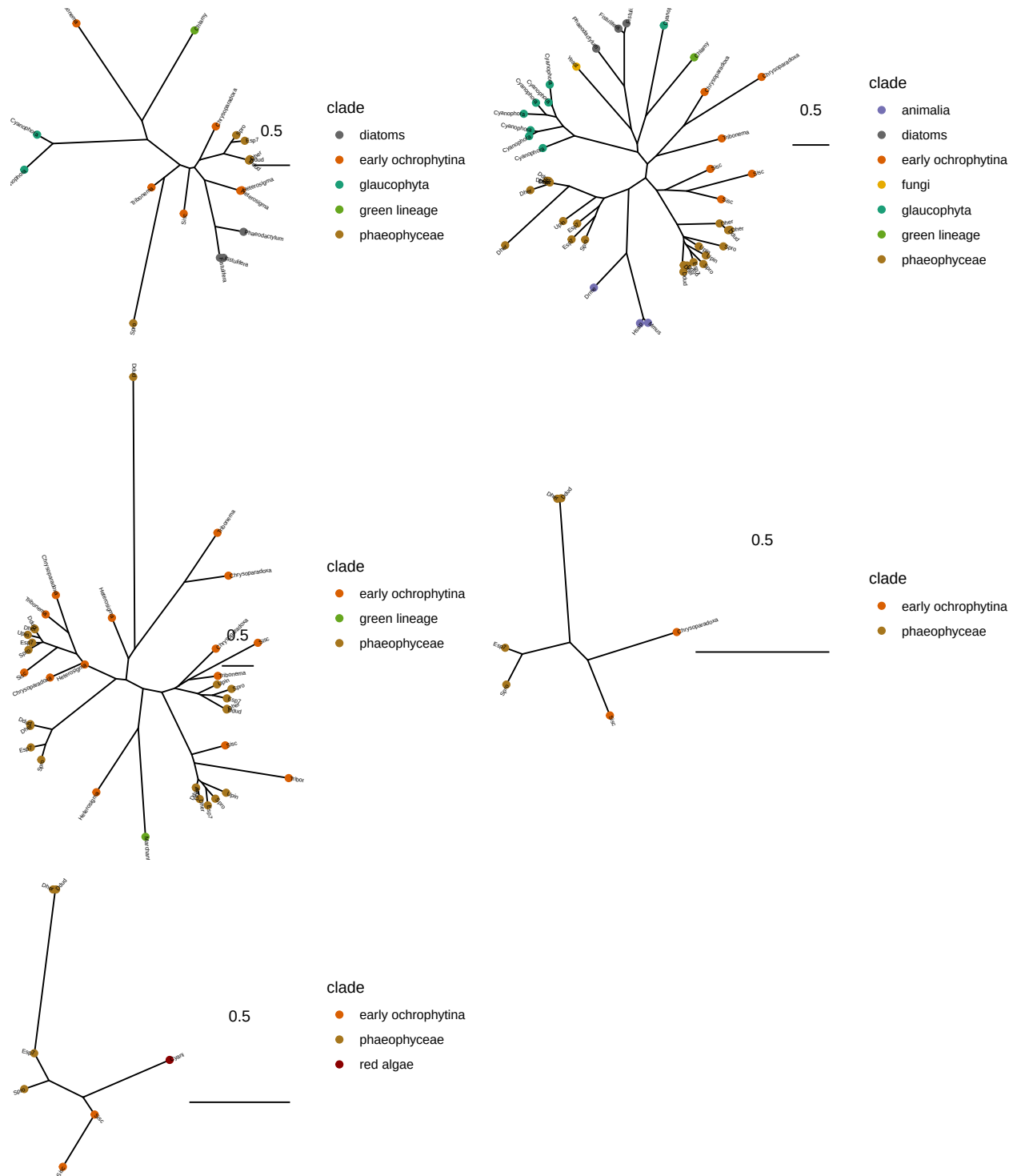
```
# ggplot2::ggsave("figure/DOT1.4_tree_fil3.pdf", width = 6, height = 4)
p5 + ggplot2::labs(title = "DOT1.5 (OG0007957)")
```

DOT1.5 (OG0007957)



```
# ggplot2::ggsave("figure/DOT1.5_tree_fil3.pdf", width = 6, height = 4)
```

```
library(patchwork)
p1 + p2 + p3 + p4 + p5 + plot_layout(ncol = 2)
```

```
# ggplot2::ggsave("figure/DOT1_all_trees_fil3.pdf", width = 10, height = 12)
```

But top blast hits are with other brown algae.

plot gene structure from interproscan

all OGs at once

Read in newick file and use ggtree to visualize

```
file_allOG <- "data/genetree/OG_all_filt3.aln.clipkit.treefile"

tree_allOG <- ggtree::read.tree(file_allOG)
# # filter tip labels to only those in Orthogroup_filt.long
tree_allOG <- treeio::drop.tip(tree_allOG,
                              setdiff(tree_allOG$tip.label,
                                       Orthogroup_filt.long$gene))
```

Plot trees via daylight method

```
# "#1B9E77" "#D95F02" "#7570B3" "#E7298A" "#66A61E" "#E6AB02" "#A6761D" "gray40"

dd = as.data.frame(Orthogroup_filt.long |>
  # replace gene_short with "" unless species = "Esp7"
  dplyr::mutate(
    gene_short = dplyr::case_when(
      species_short == "Esp7" ~ gene_short,
      TRUE ~ ""
    )
  ) |>
  dplyr::distinct(gene, gene_short, species, species_short, clade, Orthogroup))
# set consistent colour for clade points and labels

colour_code <- c(
  "early ochrophytina" = "#D95F02",
  "phaeophyceae" = "#A6761D",
  "animalia" = "#7570b3",
  "red algae" = "darkred",
  "green lineage" = "#66a61e",
  "fungi" = "#e6ab02",
  "others" = "gray40"
)

# colour by species_short
p_all <- ggtree::ggtree(tree_allOG |> phytools::midpoint_root(), layout="radial") %<+% dd +
  geom_tippoint(aes(colour = clade, label = clade),
    size=2) +
  geom_tiplab(aes(fill = clade,
    label = paste0(species_short, "\n", gene_short)),
    size=1.5,
    color = "black", # color for label font
    geom = "text",
    show.legend = FALSE) +
  ggplot2::scale_colour_manual(values = colour_code) +
  geom_treescale(width = 0.5)

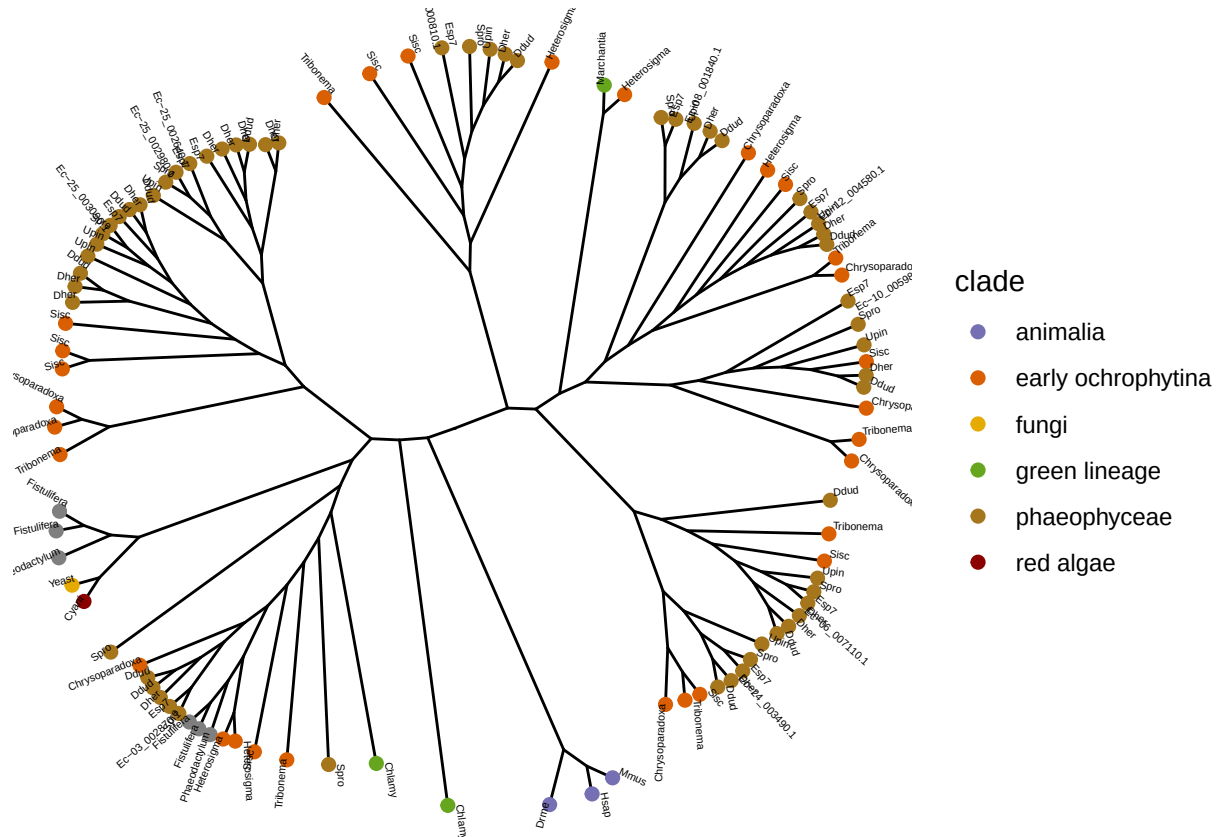
p_all <- ggtree::ggtree(tree_allOG |> phytools::midpoint_root(), layout="daylight", branch.length = "no")
  geom_tippoint(aes(colour = clade, label = clade),
    size=2) +
  geom_tiplab(aes(fill = clade,
    label = paste0(species_short, "\n", gene_short)),
    size=1.5,
```

```

color = "black", # color for label font
geom = "text",
show.legend = FALSE) +
ggplot2::scale_colour_manual(values = colour_code)
# geom_treescale(width = 0.5)

```

p_all



```

# ggplot2::ggsave("figure/DOT1_all_OG_tree_filt3.pdf", width = 8, height = 5)

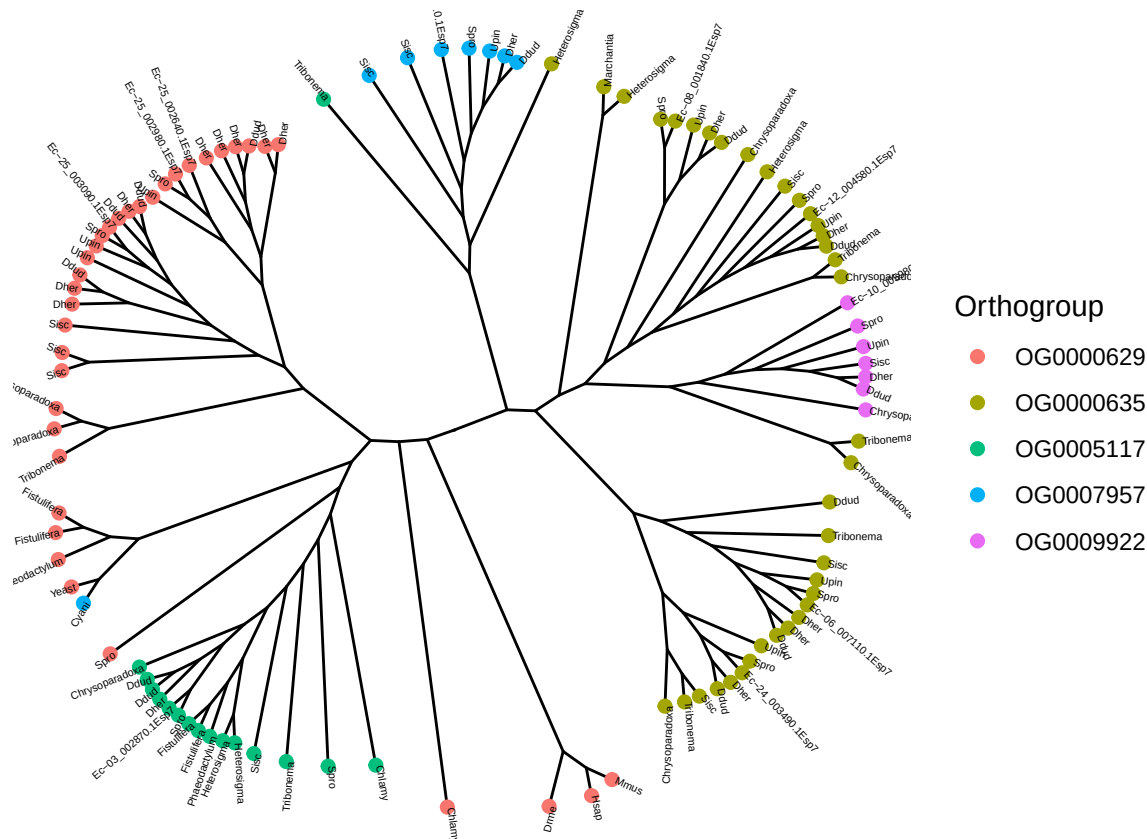
```

```

p_all <- ggtree::ggtree(tree_allOG |> phytools::midpoint_root(), layout="daylight", branch.length = "none",
  geom_tiptpoint(aes(colour = Orthogroup, label = clade),
    size=2) +
  geom_tiplab(aes(fill = Orthogroup,
    label = paste0(gene_short, species_short)),
    size=1.5,
    color = "black", # color for label font
    geom = "text",
    show.legend = FALSE)

```

p_all



```
# ggplot2::ggsave("figure/DOT1_all_OG_tree_filt3_OGonly.pdf", width = 8, height = 5)
```

gene structure with gggenes

```
library(gggenes)

domains <- readr::read_tsv("data/genetree/OG_all_filt3.fa.tsv",
                           col_names = c(
                             "protein_id", "analysis_id", "length",
                             "db", "db_accession", "description",
                             "start", "end",
                             "evalue", "status", "date",
                             "ipr_id", "ipr_description",
                             "go_terms", "pathways"
                           ))

domains_clean <- domains |>
  dplyr::select(
    protein_id, length, db, start, end, description,
    ipr_id, ipr_description
  )

Ec_gene_info <- domains_clean |>
  dplyr::filter(stringr::str_detect(protein_id, "Ec-")) |>
  dplyr::distinct(protein_id, length)

Ec_domains <- domains_clean |>
```

```

dplyr::filter(stringr::str_detect(protein_id, "Ec-")) |>
dplyr::filter(db %in% c("Pfam", "Coils", "ProSitePatterns")) |>
dplyr::mutate(
  xsubmin = start,
  xsubmax = end,
  subgene = description
) |>
dplyr::full_join(Ec_gene_info, by = c("protein_id", "length")) |>
dplyr::left_join(Orthogroup_filt.long |>
  dplyr::select(gene, Orthogroup),
  by = c("protein_id" = "gene")) |>
dplyr::mutate(protein_id_short = stringr::str_replace(protein_id, "\\.[0-9]+$", ""))

# add "Histone methylation protein DOT1" to start 1195 and end 1253 for Ec-25_002640.1 for Ec_domains
Ec_domains_add <-
  Ec_domains |>
  dplyr::filter(protein_id == "Ec-25_002640.1") |>
  dplyr::bind_rows(
    tibble(
      protein_id = "Ec-25_002640.1",
      analysis_id = NA,
      length = 1301,
      db = "Manual",
      db_accession = NA,
      description = "Histone methylation protein DOT1",
      start = 1195,
      end = 1253,
      evalue = NA,
      status = NA,
      date = NA,
      ipr_id = NA,
      ipr_description = NA,
      go_terms = NA,
      pathways = NA,
      protein_id_short = "Ec-25_002640",
      Orthogroup = "OG0000629"
    ) |>
    dplyr::mutate(
      xsubmin = start,
      xsubmax = end,
      subgene = description)
  )

# Base gene features (full-length proteins)
p <- Ec_domains |>
  dplyr::distinct(protein_id_short, length, Orthogroup) |>
  ggplot2::ggplot(
    ggplot2::aes(
      xmin = 1,
      xmax = length,
      y = protein_id_short
    )
  ) +

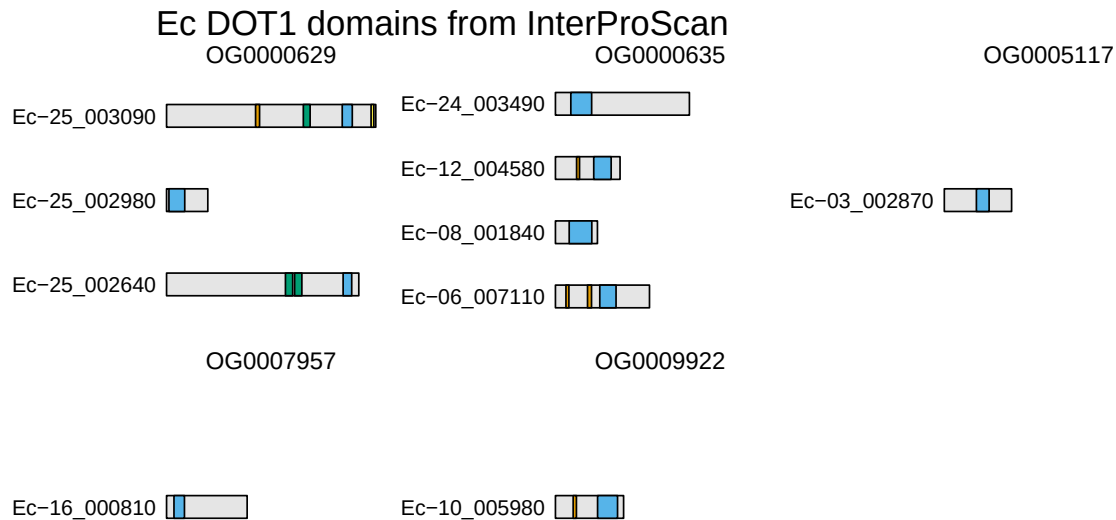
```

```

gggenes::geom_gene_arrow(
  arrowhead_height = grid::unit(0, "mm"),
  arrowhead_width = grid::unit(0, "mm"),
  fill = "grey90",
  colour = "black"
) +
gggenes::theme_genes() +
ggplot2::theme_void() +
ggplot2::labs(
  title = "Ec DOT1 domains from InterProScan"
)

# Add domains as subgenes
p +
gggenes::geom_subgene_arrow(
  data = Ec_domains |> dplyr::full_join(Ec_domains_add),
  ggplot2::aes(
    xsubmin = xsubmin,
    xsubmax = xsubmax,
    fill = subgene
  ),
  arrowhead_height = grid::unit(0, "mm"),
  arrowhead_width = grid::unit(0, "mm"),
  colour = "black"
) +
ggplot2::theme(
  legend.position = "bottom",
  # add gene labels
  axis.text.y = ggplot2::element_text(size = 8)
) +
ggplot2::guides(
  fill = ggplot2::guide_legend(
    title = "Domains",
    nrow = 2,
    byrow = TRUE
  )
) +
# use a colour blind friendly palette from paletteer
paletteer::scale_fill_paletteer_d("colorblindr::OkabeIto") +
# facet wrap based on Orthogroup_filt.long
ggplot2::facet_wrap(~Orthogroup, scales = "free_y")

```



```
# ggplot2::ggsave("figure/Ec_DOT1_domain_structure.pdf", width = 8, height = 4)
```

Extract the amino acid residues that make up the “Histone methylation protein DOT1” in Ec-25_003090.1

```
library(Biostrings)
dot1_all_seq <- Biostrings::readAAStringSet("data/genetree/OG_all_filt3.fa")

dot1_ec_seq <- dot1_all_seq["Ec-25_003090.1"]
# filter for "Histone methylation protein DOT1" from Ec_domains
dot1_ec_domain <- Ec_domains |>
  dplyr::filter(protein_id == "Ec-25_003090.1") |>
  dplyr::filter(description == "Histone methylation protein DOT1") |>
  dplyr::select(start, end)
dot1_ec_residues <- Biostrings::subseq(
  dot1_ec_seq[["Ec-25_003090.1"]],
  start = dot1_ec_domain$start,
  end = dot1_ec_domain$end
) |> AAStringSet()
# add name Ec-25_003090_dot1
names(dot1_ec_residues) <- "Ec-25_003090_dot1"

dot1_ec_seq_2 <- Biostrings::readAAStringSet("data/msc/Ec-25_002640.fasta")

dot1_ec_seq_3 <- Biostrings::readAAStringSet("data/msc/Ec-25_002980.fasta")
# filter for "Histone methylation protein DOT1" from Ec_domains
dot1_ec_domain_3 <- Ec_domains |>
  dplyr::filter(protein_id == "Ec-25_002980.1") |>
  dplyr::filter(description == "Histone methylation protein DOT1") |>
  dplyr::select(start, end)
dot1_ec_residues_3 <- Biostrings::subseq(
```

```

    dot1_ec_seq_3,
    start = dot1_ec_domain_3$start,
    end = dot1_ec_domain_3$end
) |> AAStringSet()
# add name Ec-25_002980_dot1
names(dot1_ec_residues_3) <- "Ec-25_002980_dot1"

library(msa)
all_seqs <- c(dot1_ec_seq_2, dot1_ec_residues, dot1_ec_residues_3)
alignment <- msa(all_seqs, method = "ClustalOmega")

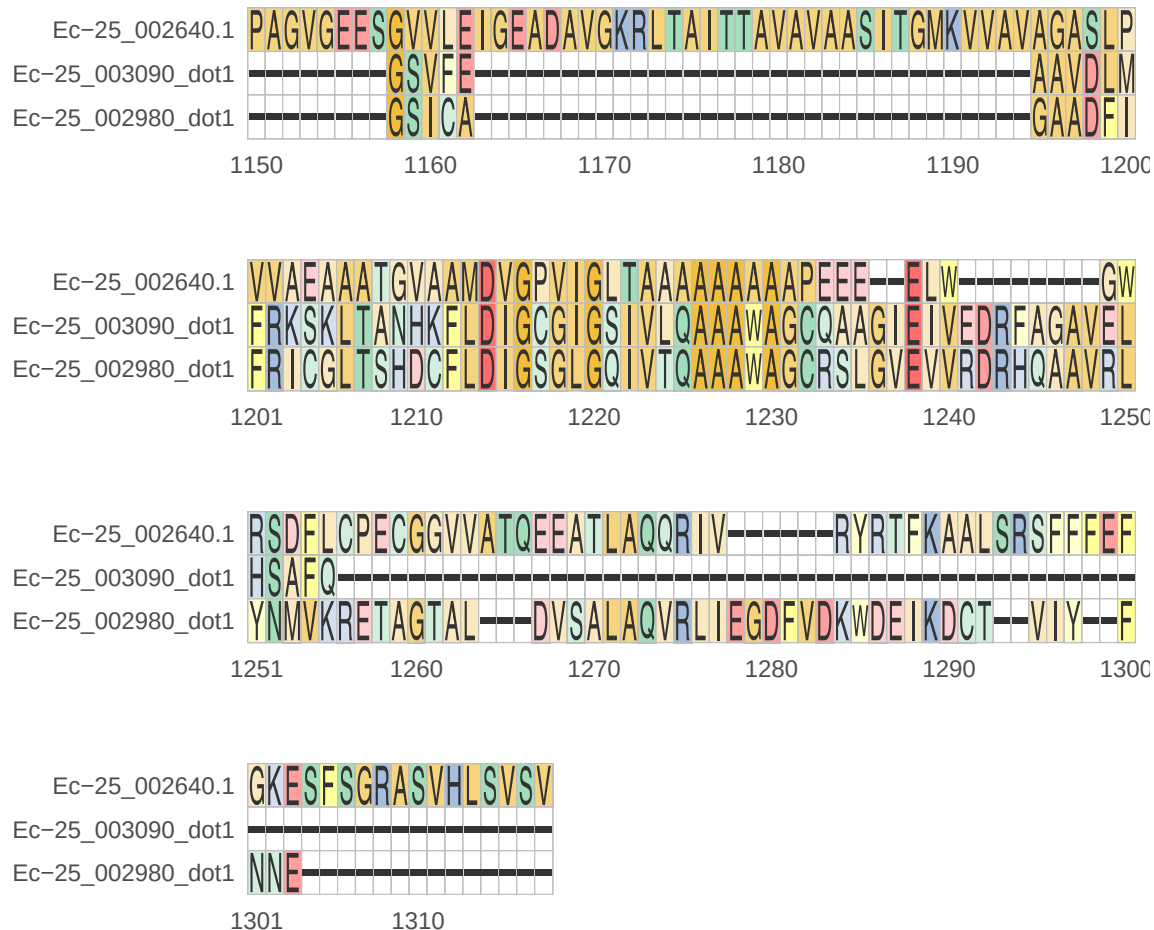
## using Gonnet
# view in R! (not save)

library(ggmsa)

alignment.processed <- alignment |> AAStringSet()

ggmsa::ggmsa(alignment.processed, start = 1150, end = Inf,
  # font = NULL,
  by_conservation = TRUE,
  seq_name = TRUE
  #start = 1189, end = 1257
) +
  ggmsa::facet_msa(field = 50)

```

```
# geom_seqlogo()
```

Now time for genes beginning with “Ddudresnayi_” (for species “Ddud”)

```
Ddud_gene_info <- domains_clean |>
  dplyr::filter(stringr::str_detect(protein_id, "Ddudresnayi_")) |>
  dplyr::distinct(protein_id, length)
Ddud_domains <- domains_clean |>
  dplyr::filter(stringr::str_detect(protein_id, "Ddudresnayi_")) |>
  dplyr::filter(db %in% c("Pfam", "Coils", "ProSitePatterns")) |>
  dplyr::mutate(
    xsubmin = start,
    xsubmax = end,
    subgene = description
  ) |>
  dplyr::full_join(Ddud_gene_info, by = c("protein_id", "length")) |>
  dplyr::left_join(Orthogroup_filt.long |>
    dplyr::select(gene, Orthogroup),
    by = c("protein_id" = "gene")) |>
  dplyr::mutate(protein_id_short = stringr::str_replace(protein_id, "\\.[0-9]+$", ""))

# Base gene features (full-length proteins)
p_ddud <- Ddud_domains |>
  dplyr::distinct(protein_id_short, length, Orthogroup) |>
  ggplot2::ggplot(
```

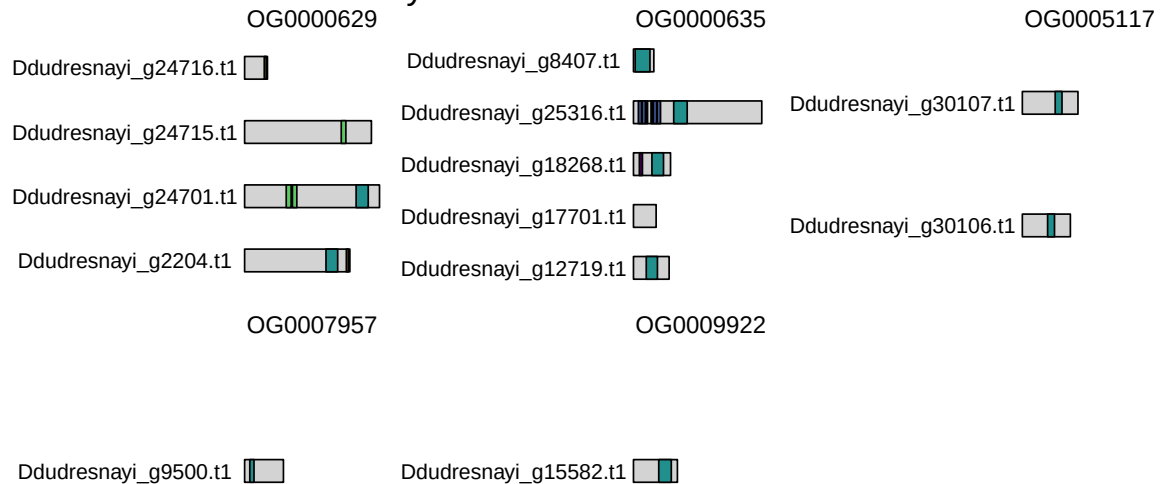
```

        ggplot2::aes(
            xmin = 1,
            xmax = length,
            y = protein_id_short
        )
    ) +
    gggenes::geom_gene_arrow(
        arrowhead_height = grid::unit(0, "mm"),
        arrowhead_width = grid::unit(0, "mm"),
        fill = "lightgray",
        colour = "black"
    ) +
    gggenes::theme_genes() +
    ggplot2::theme_void() +
    ggplot2::labs(
        title = "Ddudresnayi DOT1 domains from InterProScan"
    )
)

# Add domains as subgenes
p_ddud +
  gggenes::geom_subgene_arrow(
    data = Ddud_domains,
    ggplot2::aes(
        xsubmin = xsubmin,
        xsubmax = xsubmax,
        fill = subgene
    ),
    arrowhead_height = grid::unit(0, "mm"),
    arrowhead_width = grid::unit(0, "mm"),
    colour = "black"
  ) +
  ggplot2::theme(
    legend.position = "bottom",
    # add gene labels
    axis.text.y = ggplot2::element_text(size = 8)
  ) +
  ggplot2::guides(
    fill = ggplot2::guide_legend(
        title = "Domains",
        nrow = 2,
        byrow = TRUE
    )
  ) +
  # fill dark palette colours
  ggplot2::scale_fill_viridis_d() +
  # facet wrap based on Orthogroup_filt.long
  ggplot2::facet_wrap(~Orthogroup, scales = "free_y")

```

Ddudresnayi DOT1 domains from InterProScan



Domains

- Coil
- EsV-1-7 cysteine-rich motif
- Histone methylation
- PHD-finger
- Zinc finger RanBP2-type signature
- NA

Now for “S-ischiensis” (for species “Sisc”)

```
Sisc_gene_info <- domains_clean |>
  dplyr::filter(stringr::str_detect(protein_id, "S-ischiensis_")) |>
  dplyr::distinct(protein_id, length)
Sisc_domains <- domains_clean |>
  dplyr::filter(stringr::str_detect(protein_id, "S-ischiensis_")) |>
  dplyr::filter(db %in% c("Pfam", "Coils", "ProSitePatterns")) |>
  dplyr::mutate(
    xsubmin = start,
    xsubmax = end,
    subgene = description
  ) |>
  dplyr::full_join(Sisc_gene_info, by = c("protein_id", "length")) |>
  dplyr::left_join(Orthogroup_filt.long |>
    dplyr::select(gene, Orthogroup),
    by = c("protein_id" = "gene")) |>
  dplyr::mutate(protein_id_short = stringr::str_replace(protein_id, "\\.[0-9]+$", ""))
# Base gene features (full-length proteins)
p_sisc <- Sisc_domains |>
  dplyr::distinct(protein_id_short, length, Orthogroup) |>
  ggplot2::ggplot(
    ggplot2::aes(
      xmin = 1,
      xmax = length,
      y = protein_id_short
    )
  ) +
  gggenes::geom_gene_arrow(
    arrowhead_height = grid::unit(0, "mm"),
```

```

        arrowhead_width = grid::unit(0, "mm"),
        fill = "lightgray",
        colour = "black"
    ) +
    gggenes::theme_genes() +
    ggplot2::theme_void() +
    ggplot2::labs(
        title = "S-ischiensis DOT1 domains from InterProScan"
    )

# Add domains as subgenes
p_sisc +
    gggenes::geom_subgene_arrow(
        data = Sisc_domains,
        ggplot2::aes(
            xsubmin = xsubmin,
            xsubmax = xsubmax,
            fill = subgene
        ),
        arrowhead_height = grid::unit(0, "mm"),
        arrowhead_width = grid::unit(0, "mm"),
        colour = "black"
    ) +
    ggplot2::theme(
        legend.position = "bottom",
        # add gene labels
        axis.text.y = ggplot2::element_text(size = 8)
    ) +
    ggplot2::guides(
        fill = ggplot2::guide_legend(
            title = "Domains",
            nrow = 2,
            byrow = TRUE
        )
    ) +
    # fill dark palette colours
    ggplot2::scale_fill_viridis_d() +
    # facet wrap based on Orthogroup_filt.long
    ggplot2::facet_wrap(~Orthogroup, scales = "free_y")

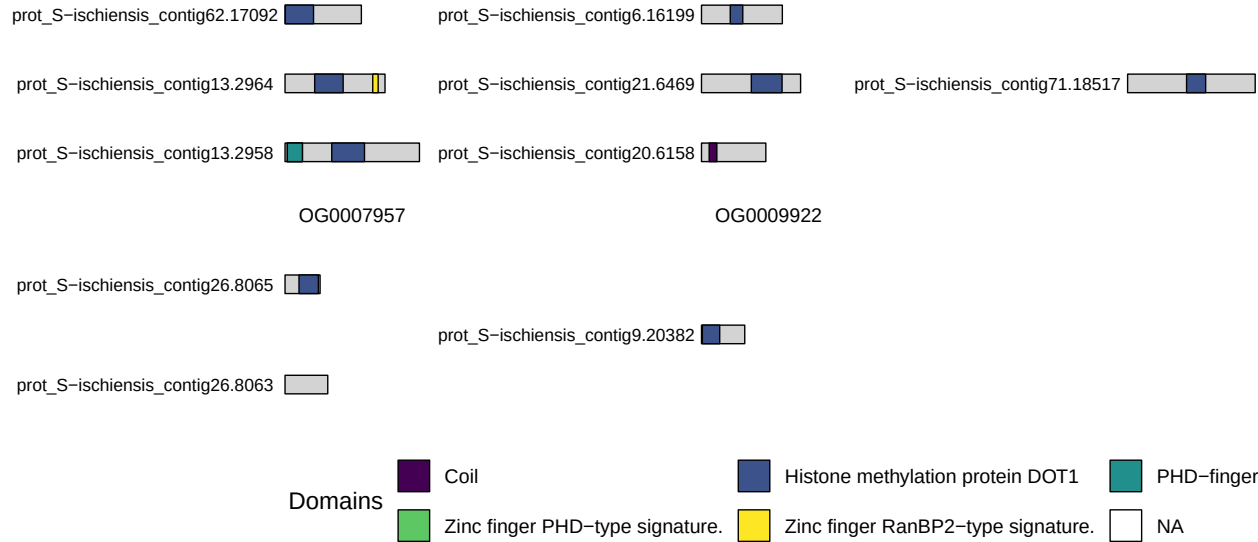
```

S-ischiensis DOT1 domains from InterProScan

OG0000629

OG0000635

OG0005117



All species at once?

```
library(gggenes)
```

```
all_browns_geneIDs <-
  Orthogroup_filt.long |>
  dplyr::filter(species %in% c("Esp7", "Ddud", "Dher", "Upin", "Spro", "Sisc")) |>
  dplyr::select(gene, species, gene_short)
```

```
all_browns_gene_info <- domains_clean |>
  dplyr::mutate(protein_id = stringr::str_replace(protein_id, "ori_\\+", "ori_")) |>
  dplyr::filter(protein_id %in% all_browns_geneIDs$gene) |>
  # dplyr::filter(stringr::str_detect(protein_id, "Ec-/Ddudresnayi_/prot_D-herbacea_/UNPIN/S-prom
  dplyr::distinct(protein_id, length)
  # remove from protein_id "\\t1", "prot_", "ISG\\|", anything after "\\|len_" and call the new
  # dplyr::mutate(protein_id_short = protein_id |>
  #   stringr::str_replace("prot_", "") |>
  #   stringr::str_replace("ISG\\|", "") |>
  #   stringr::str_replace("\\t1$", "") |>
  #   stringr::str_replace("\\|len_.*$", ""))
```

```
all_browns_domains <- domains_clean |>
  dplyr::mutate(protein_id = stringr::str_replace(protein_id, "ori_\\+", "ori_")) |>
  dplyr::filter(protein_id %in% all_browns_geneIDs$gene) |>
  dplyr::filter(db %in% c("Pfam", "Coils", "ProSitePatterns")) |>
  dplyr::mutate(
    xsubmin = start,
    xsubmax = end,
    subgene = description
  ) |>
  dplyr::full_join(all_browns_gene_info, by = c("protein_id", "length")) |>
  dplyr::left_join(Orthogroup_filt.long |>
    dplyr::select(gene, Orthogroup),
    by = c("protein_id" = "gene")) |>
```

```

dplyr::mutate(protein_id_short = protein_id |>
  stringr::str_replace("prot_", "") |>
  stringr::str_replace("ISG\\|", "") |>
  stringr::str_replace("\\.t1$", "") |>
  stringr::str_replace("\\.1$", "") |>
  stringr::str_replace("\\|len_.*$", "")) |>
# as factor for subgene beginning with "Coil", "Histone methylation protein DOT1", "PHD-finger"
dplyr::mutate(
  subgene = factor(subgene,
    levels = c(
      "Coil",
      "Histone methylation protein DOT1",
      "PHD-finger",
      "Zinc finger RanBP2-type signature.",
      "Zinc finger PHD-type signature.",
      "EsV-1-7 cysteine-rich motif"
    ),
    ordered = TRUE
  )
) |>
# "Zinc finger PHD-type signature." overlaps with "PHD-finger", so remove it
dplyr::filter(subgene != "Zinc finger PHD-type signature.")

# Base gene features (full-length proteins)
p <- all_browns_domains |>
dplyr::distinct(protein_id_short, length, Orthogroup) |>
ggplot2::ggplot(
  ggplot2::aes(
    xmin = 1,
    xmax = length,
    y = protein_id_short
  )
) +
gggenes::geom_gene_arrow(
  arrowhead_height = grid::unit(0, "mm"),
  arrowhead_width = grid::unit(0, "mm"),
  fill = "grey90",
  colour = "black"
) +
gggenes::theme_genes() +
ggplot2::theme_void() +
ggplot2::labs(
  title = "all brown + Sisc DOT1 domains from InterProScan"
)

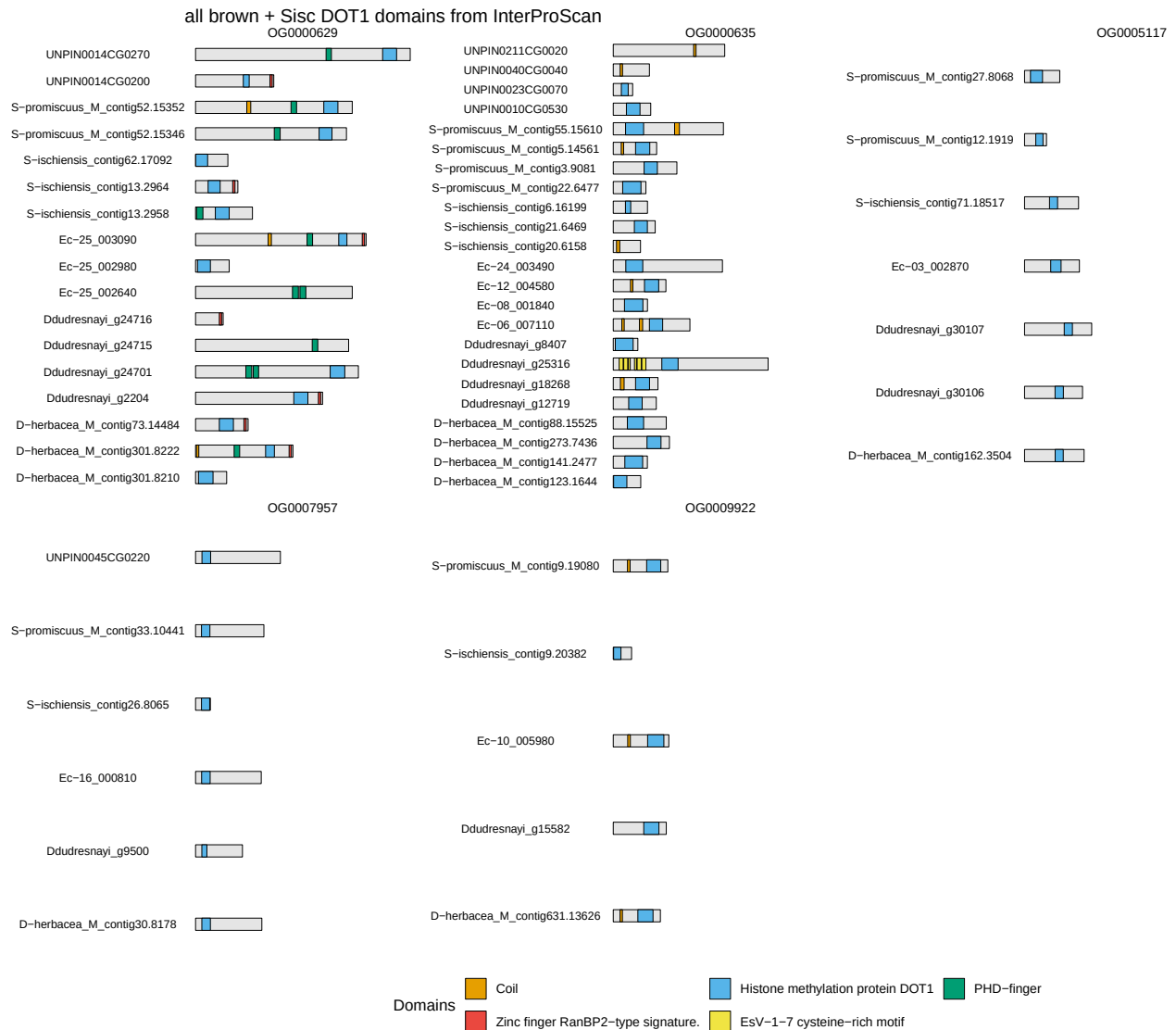
# Add domains as subgenes
p +
gggenes::geom_subgene_arrow(
  data = all_browns_domains |> tidyr::drop_na(),
  ggplot2::aes(
    xsubmin = xsubmin,
    xsubmax = xsubmax,
    fill = subgene
  )
)

```

```

    ),
    arrowhead_height = grid::unit(0, "mm"),
    arrowhead_width = grid::unit(0, "mm"),
    colour = "black"
  ) +
  ggplot2::theme(
    legend.position = "bottom",
    # add gene labels
    axis.text.y = ggplot2::element_text(size = 8)
  ) +
  ggplot2::guides(
    fill = ggplot2::guide_legend(
      title = "Domains",
      nrow = 2,
      byrow = TRUE
    )
  ) +
  # use a colour blind friendly palette from paletteer
  # paletteer::scale_fill_paletteer_d("colorblindr::OkabeIto") +
  ggplot2::scale_fill_manual(values = c(
    "Coil" = "#E69F00FF",
    "Histone methylation protein DOT1" = "#56B4E9FF",
    "PHD-finger" = "#009E73FF",
    "Zinc finger RanBP2-type signature." = "#ea483f",
    "EsV-1-7 cysteine-rich motif" = "#F0E442FF"
  )) +
  # facet wrap based on Orthogroup_filt.long
  ggplot2::facet_wrap(~Orthogroup, scales = "free_y")

```



```
# ggplot2::ggsave("figure/all_browns_DOT1_domain_structure.pdf", width = 12, height = 10)
# c("#E69F00FF", "#56B4E9FF", "#009E73FF", "#F0E442FF", "#0072B2FF", "#D55E00FF", "#CC79A7FF", "#999999FF", "#ea483f")
```

```
devtools::session_info(pkgs = "attached")
```

```
## - Session info -----
## setting value
## version R version 4.4.0 (2024-04-24)
## os macOS 15.7.2
## system x86_64, darwin20
## ui X11
## language (EN)
## collate en_US.UTF-8
## ctype en_US.UTF-8
## tz Europe/Berlin
## date 2026-01-30
```



```
## pandoc 3.4 @ /Applications/RStudio.app/Contents/Resources/app/quarto/bin/tools/x86_64/ (via rmarkdown)
##
## - Packages -----
## package      * version date (UTC) lib source
## BiocGenerics * 0.50.0 2024-04-30 [1] Bioconductor 3.19 (R 4.4.0)
## Biostrings   * 2.72.1 2024-06-02 [1] Bioconductor 3.19 (R 4.4.0)
## dplyr         * 1.1.4 2023-11-17 [1] CRAN (R 4.4.0)
## forcats      * 1.0.0 2023-01-29 [1] CRAN (R 4.4.0)
## GenomeInfoDb * 1.40.1 2024-05-24 [1] Bioconductor 3.19 (R 4.4.0)
## gggenes      * 0.5.1 2023-09-05 [1] CRAN (R 4.4.0)
## ggmsa        * 1.17.0 2025-11-19 [1] Github (YuLab-SMU/ggmsa@4f1048b)
## ggplot2      * 4.0.0 2025-09-11 [1] CRAN (R 4.4.1)
## ggtree       * 4.1.1 2025-11-17 [1] Github (YuLab-SMU/ggtree@57a3e6b)
## IRanges      * 2.38.1 2024-07-03 [1] Bioconductor 3.19 (R 4.4.1)
## lubridate    * 1.9.4 2024-12-08 [1] CRAN (R 4.4.1)
## msa          * 1.36.1 2024-07-24 [1] Bioconductor 3.19 (R 4.4.1)
## patchwork    * 1.3.2 2025-08-25 [1] CRAN (R 4.4.1)
## purrr        * 1.1.0 2025-07-10 [1] CRAN (R 4.4.1)
## readr        * 2.1.5 2024-01-10 [1] CRAN (R 4.4.0)
## S4Vectors    * 0.42.1 2024-07-03 [1] Bioconductor 3.19 (R 4.4.1)
## stringr      * 1.5.2 2025-09-08 [1] CRAN (R 4.4.1)
## tibble       * 3.3.0 2025-06-08 [1] CRAN (R 4.4.1)
## tidyr        * 1.3.1 2024-01-24 [1] CRAN (R 4.4.0)
## tidyverse    * 2.0.0 2023-02-22 [1] CRAN (R 4.4.0)
## XVector      * 0.44.0 2024-04-30 [1] Bioconductor 3.19 (R 4.4.0)
##
## [1] /Library/Frameworks/R.framework/Versions/4.4-x86_64/Resources/library
##
## -----
```